

# OPERATING SYSTEMS

---

## Memory Management

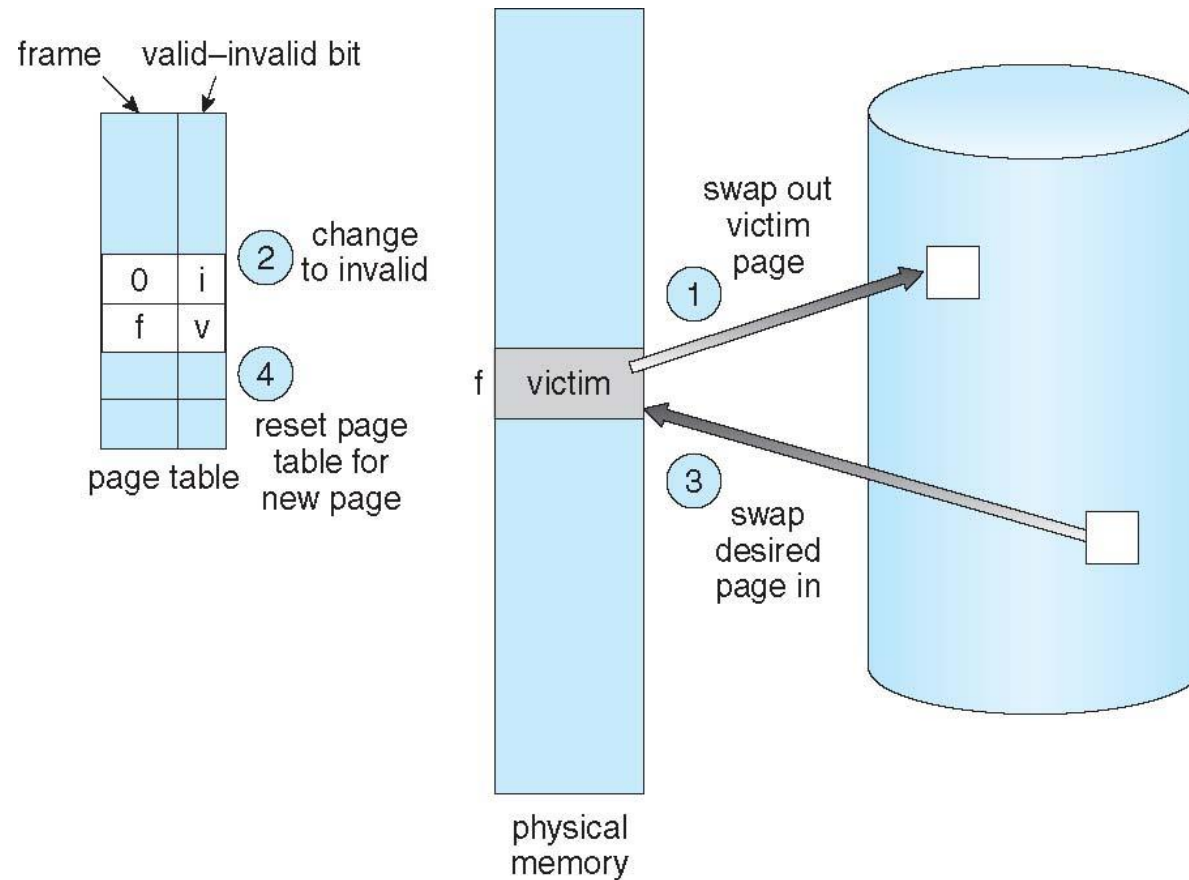
**S.Thenmozhi**

Department of Computer Applications

### What happens if there is no free frame?

- Page replacement – find some page in memory, but not really in use, page it out
  - Use an algorithm which will result in minimum number of page faults
- Same page may be brought into memory several times

- Steps for page replacement algorithm
  - Select the victim frame in the physical memory
  - Swap out the victim page
  - Swap in the desired page
  - Update the page table



### FIFO Replacement Algorithm

The oldest page is in the front of the queue is selected for removal

Following is the reference string made by the CPU

Assume that the number of frames in the main memory is 3

reference string

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

### FIFO Replacement Algorithm

reference string

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

|   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 7 | 7 | 7 | 2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   | 0 | 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|   |   | 1 | 1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

|   |   |   |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|---|---|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 2 | 2 | 4 | 4 | 4 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 3 | 3 | 3 | 2 | 2 | 2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 1 | 0 | 0 | 0 | 3 | 3 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

|   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 1 | 1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 3 | 2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

|   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---|---|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 7 | 7 | 7 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 1 | 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 2 | 2 | 1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

page frames

No of Page faults: 15

Rate of Page fault: 75%

### FIFO Replacement Algorithm

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 2 | 1 | 5 | 6 | 2 | 1 | 2 | 3 | 7 | 6 | 3 |
| 1 | 1 | 1 | 4 | 4 | 4 | 4 | 6 | 6 | 6 | 6 | 3 | 3 | 3 | 3 |
|   | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 7 | 7 | 7 |
|   |   | 3 | 3 | 3 | 3 | 5 | 5 | 5 | 1 | 1 | 1 | 1 | 6 | 6 |
| F | F | F | F | H | F | F | F | F | F | H | F | F | F | H |

No of Page faults: 12

Rate of Page fault: 80%



# THANK YOU

---

**S. Thenmozhi**

Department of Computer Applications

**[thenmozhis@pes.edu](mailto:thenmozhis@pes.edu)**

+91 80 6666 3333 Extn 393